Jiawei Huang . Jarvis Consulting

Data Engineer and aspiring Mainframe Developer with a background in Actuarial Science and Financial Accountability. Experienced in building data pipelines and financial systems, with hands-on COBOL development skills including indexed file I/O, VSAM architecture, JCL, ISPF, and modular program design. Passionate about applying enterprise mainframe technologies to the high-reliability demands of financial data processing, where legacy and modern systems intersect.

Skills

Proficient: Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, Python (Numpy, Pandas, Matplotlib), Spark, Tableau, MS Excel (formulas, conditional formatting, pivot table)

Competent: Cobol, JCL, ISPF, DB2, Databricks, Docker, Java

Familiar: SAS, R, Hadoop, Redis, Google Cloud Platform, Microsoft Azure

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_JiaweiHuang

Student Registration System [GitHub]:

- Designed and built a full-featured student records management system in COBOL, comprising nine independently compiled programs orchestrated through a central menu interface
- Implemented complete CRUD functionality against a keyed indexed file with random and sequential access patterns
- Developed a control-break report generator that groups student records by enrollment date and writes formatted output to a sequential report file
- Applied defensive file handling throughout, validating FILE STATUS codes after every I/O operation and handling duplicate key, record-not-found, and layout mismatch conditions

Cluster Monitor [GitHub]:

- Developed a Linux host monitoring agent for system administrators and DevOps, providing real-time and historical insights into cluster resource utilization.
- Designed a relational schema in PostgreSQL to support static hardware data and high-frequency time-series metrics (CPU, memory), linking host_info and host_usage tables.
- Built data collection scripts in Bash leveraging Linux kernel interfaces; automated ingestion via cron jobs.
- Ensured data accuracy by validating database records against live system outputs (vmstat).
- Containerized PostgreSQL using Docker with persistent storage (Docker volumes) and managed development workflow via GitHub.

Java Grep App [GitHub]:

- Developed a Java-based application replicating Linux grep to search regex patterns across directory trees and output matching results, supporting log analysis for large-scale systems.
- Implemented both iterative (loops) and functional (Streams, Lambda) approaches to optimize performance and code readability.
- Managed dependencies using Maven and containerized the application with Docker for consistent cross-platform deployment.
- Validated accuracy by benchmarking results against native grep outputs using sample files and log data.
- Maintained version control and development workflow using GitHub.

London Gift Shop Data Analytics [GitHub]:

- Designed an ETL pipeline for processing 1M+ rows of retail data using Jupyter Notebook
- Created interactive visualizations to communicate seasonal churn trends to stakeholders
- Developed an RFM model that classified users based on shopping patterns and designed marketing strategies for customer segmentation

Stock Data ETL Pipeline [GitHub]:

- Built a robust data ingestion engine to extract daily stock records from REST APIs into a Databricks environment
- Leveraged Apache Spark and Delta Lake to build a structured DLT pipeline, transforming raw JSON responses into high-performance analytical tables

- Visualized key financial metrics by building dashboards on refined data layers, reducing manual reporting time
- Managed job orchestration and error handling for daily data loads, ensuring 99.9% pipeline uptime

Highlighted Projects

Regression Analysis for Red Wine Quality Factors: Performed regression analysis on the red wine dataset with 1.6K observations, utilized SAS for computation. Identified and addressed outliers through Cook's distance analysis and implemented necessary tests, enhanced model stability and increased R2. Collaborated with team members to optimize the model for accurate future predictions

Professional Experiences

Big Data Developer, Bank of Suzhou (2024-2025): Developed and maintained ETL pipelines on Alibaba Cloud and MaxCompute to process and transform large-scale banking data for downstream analytics and reporting. Designed and managed data warehouse tables, partitioning strategies, and data models to support efficient data storage and retrieval. Optimized large-scale queries and data processing jobs, improving performance and reducing execution costs within the cloud data platform. Implemented data quality checks and reconciliation processes to ensure reliability and integrity of enterprise reporting data. Partnered with analysts and business stakeholders to deliver scalable data solutions supporting regulatory reporting, operational monitoring, and business intelligence initiatives.

Operational Data Specialist, OTT Financial Inc. (2021-2022): Conducted in-depth revenue analysis through customer segmentation via Python, identified specific age groups and regions with high purchasing power, resulting in strategic recommendation that bolstered sales by 8%. Designed and automated weekly reporting process by building calculation mode in Python (Pandas, NumPy). Led promotional effect analysis by conducting A/B testing and data transformations to inform and refine marketing strategies, provided recommendations for marketing team with cost-efficient discount plans to boost sales. Managed the end to end business intelligence cycle, overseeing data analysis and visualizations in Tableau dashboards while ensuring governance of merchant sales and user demographic data for future analytics.

Education

York University (2022-2023), Master of Financial Accountability, Administrative Studies - Relevant Courses: Enterprise Risk Management, Corporate Governance, Synthesis of theory and practice, governance and accountability

York University (2017-2021), Bachelor of Arts Hons in Actuarial Science, Mathematics and Statistics - Relevant Courses: The Mathematical Theory of Interest, Models of Financial Economics, Scientific Computation of Financial Application, Time Series Analysis, Regression Analysis

Miscellaneous

- Coursera IBM Data Science Professional
- Basketball player
- Poker Player
- Close-up Magician